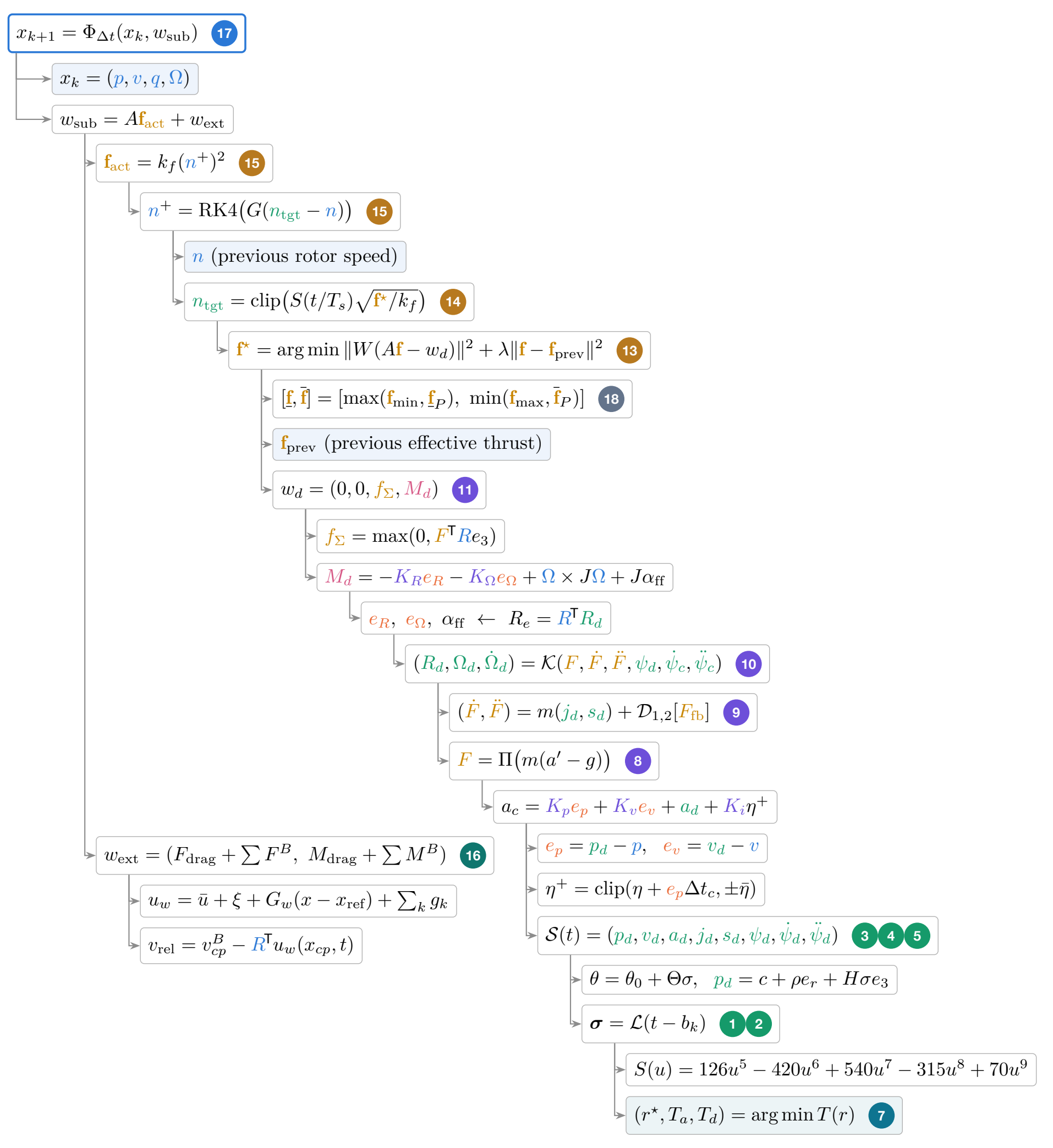
isaac-drone-sim: The Complete Math Pipeline as a Formula Logic Tree

Purely symbolic: no configuration values are substituted; every formula was checked against the `isaac_drone/` source.

Thin arrows inside a module = "obtained by substituting the upstream formula"; thick arrows = quantities passed between modules; dashed = feedback to the next step / degenerate branch.

Overall logic tree: everything x_{k+1} depends on

parent → child = "is computed from"; shaded leaves = previous-step state / planned at reset



- $x_{k+1} = \Phi_{\Delta t}(x_k, w_{\text{sub}})$ (17)
  - $x_k = (p, v, q, \Omega)$
  - $w_{\text{sub}} = A\mathbf{f}_{\text{act}} + w_{\text{ext}}$
    - $\mathbf{f}_{\text{act}} = k_f (n^+)^2$ (15)
      - $n^+ = \text{RK4}(G(n_{\text{tgt}} - n))$ (15)
        - n (previous rotor speed)
        - $n_{\text{tgt}} = \text{clip}\big(S(t/T_s)\sqrt{\mathbf{f}^\star/k_f}\big)$ (14)
          - $\mathbf{f}^\star = \arg\min \|W(A\mathbf{f} - w_d)\|^2 + \lambda\|\mathbf{f} - \mathbf{f}_{\text{prev}}\|^2$ (13)
            - $[\underline{\mathbf{f}}, \bar{\mathbf{f}}] = [\max(\mathbf{f}_{\min}, \underline{\mathbf{f}}_P),\ \min(\mathbf{f}_{\max}, \bar{\mathbf{f}}_P)]$ (18)
            - $\mathbf{f}_{\text{prev}}$ (previous effective thrust)
            - $w_d = (0, 0, f_\Sigma, M_d)$ (11)
              - $f_\Sigma = \max(0, F^\mathsf{T} R e_3)$
              - $M_d = -K_R e_R - K_\Omega e_\Omega + \Omega \times J\Omega + J\alpha_{\text{ff}}$
                - $e_R,\ e_\Omega,\ \alpha_{\text{ff}} \leftarrow\ R_e = R^\mathsf{T} R_d$
                  - $(R_d, \Omega_d, \dot\Omega_d) = \mathcal{K}(F, \dot F, \ddot F, \psi_d, \dot\psi_c, \ddot\psi_c)$ (10)
                    - $(\dot F, \ddot F) = m(j_d, s_d) + \mathcal{D}_{1,2}[F_{\text{fb}}]$ (9)
                    - $F = \Pi\big(m(a' - g)\big)$ (8)
                      - $a_c = K_p e_p + K_v e_v + a_d + K_i \eta^+$
                        - $e_p = p_d - p,\quad e_v = v_d - v$
                        - $\eta^+ = \text{clip}(\eta + e_p \Delta t_c, \pm\bar\eta)$
                        - $\mathcal{S}(t) = (p_d, v_d, a_d, j_d, s_d, \psi_d, \dot\psi_d, \ddot\psi_d)$ (3) (4) (5)
                          - $\theta = \theta_0 + \Theta\sigma,\quad p_d = c + \rho e_r + H\sigma e_3$
                          - $\boldsymbol\sigma = \mathcal{L}(t - b_k)$ (1) (2)
                            - $S(u) = 126u^5 - 420u^6 + 540u^7 - 315u^8 + 70u^9$
                            - $(r^\star, T_a, T_d) = \arg\min T(r)$ (7)
    - $w_{\text{ext}} = (F_{\text{drag}} + \sum F^B,\ M_{\text{drag}} + \sum M^B)$ (16)
      - $u_w = \bar u + \xi + G_w(x - x_{\text{ref}}) + \sum_k g_k$
      - $v_{\text{rel}} = v^B_{cp} - R^\mathsf{T} u_w(x_{cp}, t)$

Master formula: one physics step $k \to k+1$

$$x_{k+1} = \Phi_{\Delta t}\Big(x_k,\ A\,k_f \odot \Big[\mathcal{M}_{\Delta t}\Big(n_k,\ \mathcal{R}\Big(\mathcal{A}\big(\mathcal{C}\big(x_k,\ \mathcal{G}(\mathcal{L}(t_k))\big)\big),\ [\underline{\mathbf{f}}, \bar{\mathbf{f}}]\Big),\ t_k\Big)\Big)\Big]^2 + w_{\text{ext}}(x_k, t_k)\Big)$$

Layer by layer (badges on the right = module numbers)

$x_{k+1} = \Phi_{\Delta t}(x_k,\ w_{\text{sub}}),\quad w_{\text{sub}} = A\mathbf{f}_{\text{act}} + w_{\text{ext}}(x_k, t_k)$	(17) (16)
$\mathbf{f}_{\text{act}} = k_f \odot (n^+)^2,\quad n^+ = \mathcal{M}_{\Delta t}(n_k, n_{\text{tgt}})$	(15)
$n_{\text{tgt}} = \mathcal{R}(\mathbf{f}^\star, t_k) = \text{clip}\Big(S\big(\tfrac{t_k}{T_s}\big)\sqrt{\mathbf{f}^\star/k_f},\ n_{\min}, n_{\max}\Big)$	(14)
$\mathbf{f}^\star = \mathcal{A}\big(w_d;\ [\underline{\mathbf{f}}, \bar{\mathbf{f}}],\ \mathbf{f}_{\text{prev}}\big)$	(13) (18)
$w_d = \mathcal{C}\big(x_k,\ \mathcal{S}(t_k)\big)$ (recomputed only when $k \bmod N_c = 0$)	(8)–(11) or (12)
$\mathcal{S}(t) = \mathcal{G}\big(\boldsymbol\sigma(t)\big),\quad \boldsymbol\sigma = \mathcal{L}\big(t;\ T \text{ or } (r^\star, T_a, T_d)\big)$	(3)–(5) (1) (2) (7)

Notation and Legend

Colors actual p, v, R, Ω, n desired / reference $p_d, R_d, \Omega_d, n_{\text{tgt}}$ error e_p, e_R, ϱ thrust $F, f_\Sigma, \mathbf{f}$ torque M_d gain K_p, K_R

Frames world W (z up), body B ($+z$ = thrust direction); $R: B \to W$; $e_3 = (0,0,1)^\mathsf{T}$; g = gravity in W

Time t absolute sim time; Δt physics step; $\Delta t_c = N_c \Delta t$ control period; superscript $+$ = next step; $x^{(k)}$ = k-th derivative

Symbols ρ helix radius; r cruise progress rate; r_i rotor position; c CoM in body frame; f_Σ collective thrust; $\mathbf{f} = (f_1, \dots, f_4)^\mathsf{T}$

Operators $\text{clip}(x, a, b) = \min(\max(x, a), b)$; $[\cdot]$ Iverson bracket (1 if true); $\odot$ elementwise product; A^+ pseudoinverse

⟶	substitution (inside a module)	- - ->	degenerate / roll-back branch
⟶	data flow between modules (same step)	- - ->	feedback to next step / once at reset

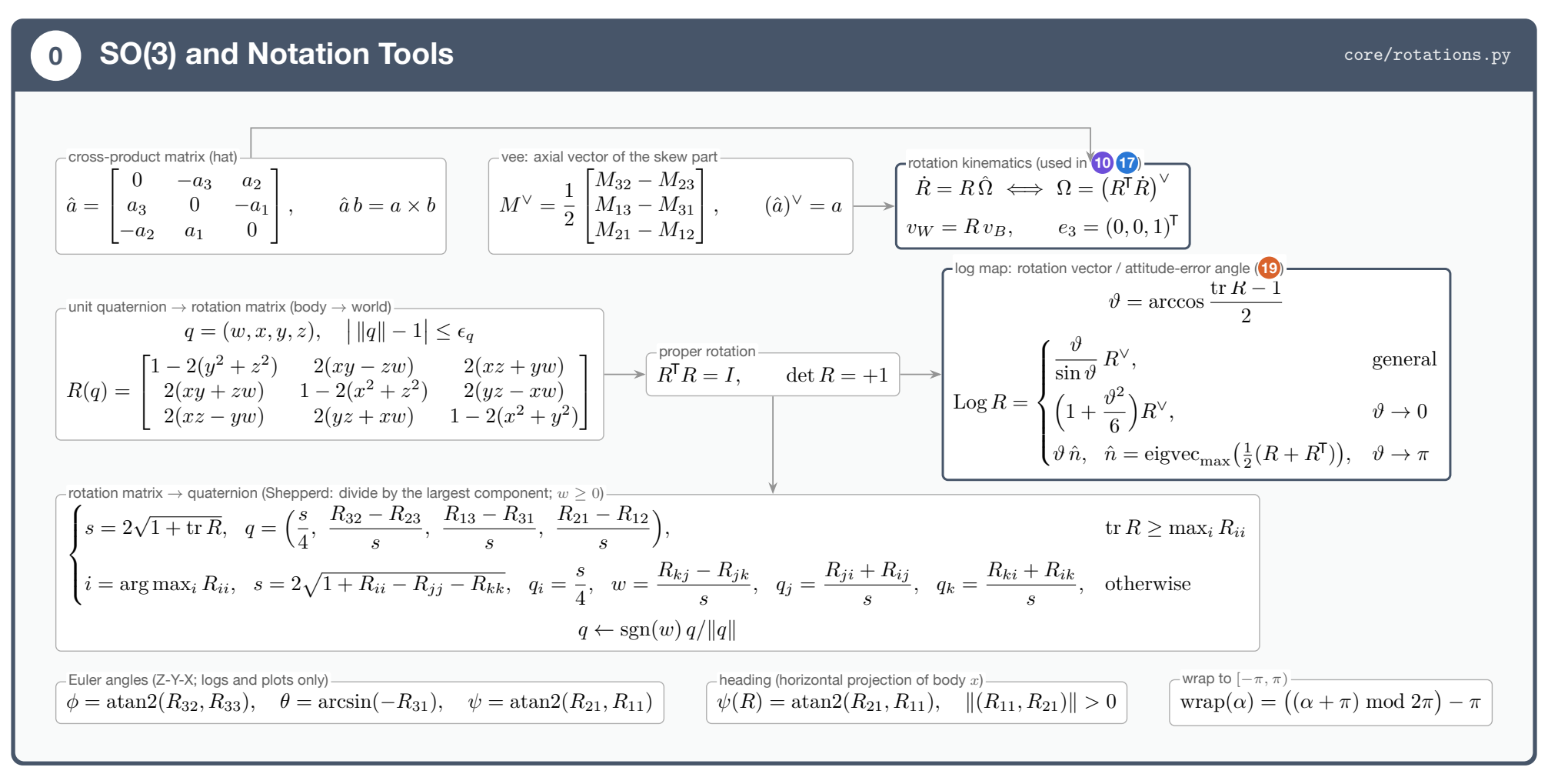
formula · module output · input / definition · condition / degenerate branch

Execution Order of One Physics Step (runtime/loop.py)

Once at reset $x_0 \to$ (3) $(p_0, \psi_0) \to$ (7) $(r^\star, T_a, T_d) \to$ (2) $\to$ (6) screen

Every step $x_k \xrightarrow{k \bmod N_c = 0} \mathcal{S}(t_k)$ (3)–(5) $\to w_d$ (8)–(11) $\to \mathbf{f}^\star$ (13) $\to n_{\text{tgt}}$ (14) $\to \ell_{\text{alloc}} \to$ (8)

$\to w_{\text{ext}}$ (16) $\to n^+, \mathbf{f}_{\text{act}}$ (15) $\to x_{k+1}$ (17) $\to$ telemetry (19) $\to$ completion (20) $\to$ battery (18)



⓪ SO(3) and Notation Tools

`core/rotations.py`

cross-product matrix (hat)
$$\hat a = \begin{bmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{bmatrix},\qquad \hat a\, b = a \times b$$

vee: axial vector of the skew part
$$M^\vee = \frac{1}{2}\begin{bmatrix} M_{32} - M_{23} \\ M_{13} - M_{31} \\ M_{21} - M_{12} \end{bmatrix},\qquad (\hat a)^\vee = a$$

rotation kinematics (used in (10) (17))
$$\dot R = R\hat\Omega \iff \Omega = (R^\mathsf{T}\dot R)^\vee,\qquad v_W = R\,v_B,\qquad e_3 = (0,0,1)^\mathsf{T}$$

unit quaternion → rotation matrix (body → world)
$$q = (w, x, y, z),\qquad \big|\,\|q\| - 1\big| \le \epsilon_q$$
$$R(q) = \begin{bmatrix} 1 - 2(y^2+z^2) & 2(xy - zw) & 2(xz + yw) \\ 2(xy + zw) & 1 - 2(x^2+z^2) & 2(yz - xw) \\ 2(xz - yw) & 2(yz + xw) & 1 - 2(x^2+y^2) \end{bmatrix}$$

proper rotation
$$R^\mathsf{T} R = I,\qquad \det R = +1$$

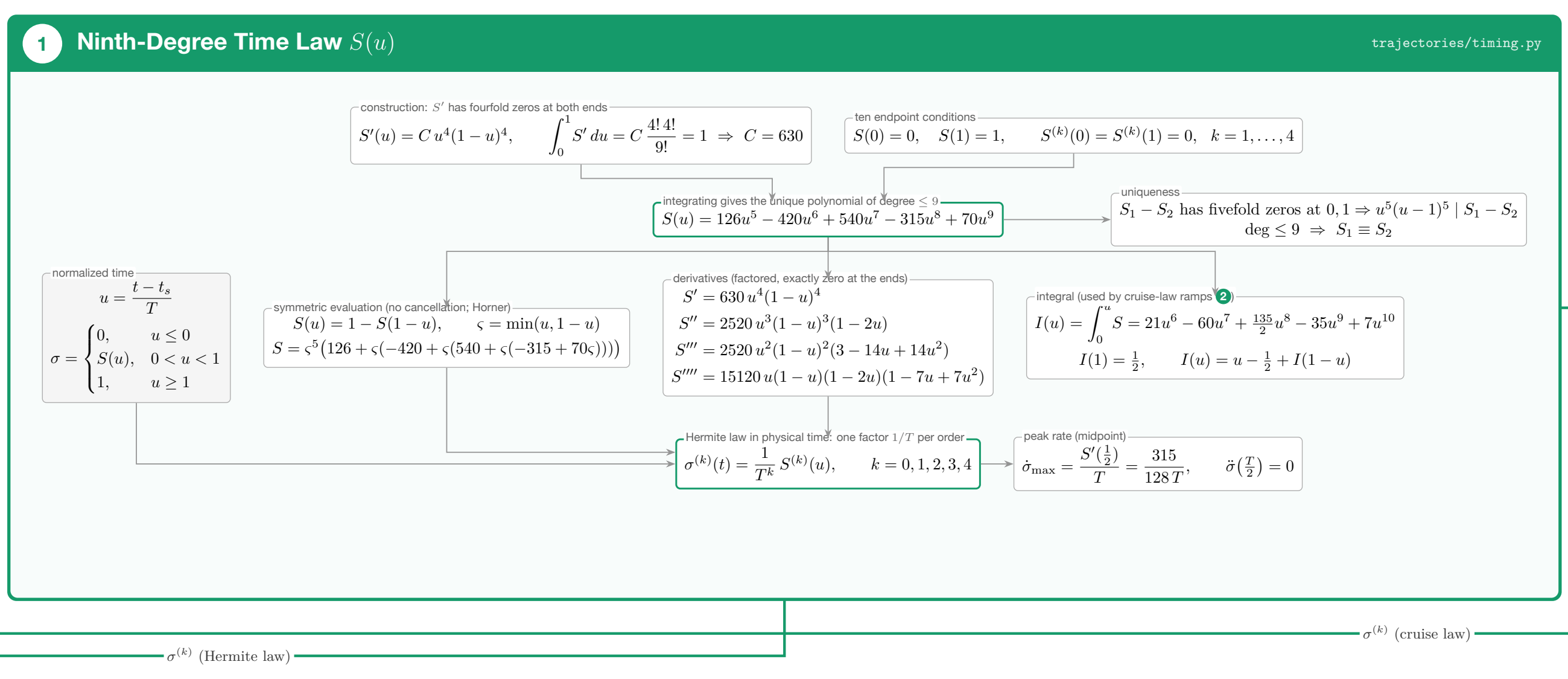
log map: rotation vector / attitude-error angle (19)
$$\vartheta = \arccos\frac{\operatorname{tr} R - 1}{2}$$
$$\operatorname{Log} R = \begin{cases} \dfrac{\vartheta}{\sin\vartheta} R^\vee, & \text{general} \\ \Big(1 + \dfrac{\vartheta^2}{6}\Big) R^\vee, & \vartheta \to 0 \\ \vartheta\,\hat n,\quad \hat n = \operatorname{eigvec}_{\max}\big(\tfrac12(R + R^\mathsf{T})\big), & \vartheta \to \pi \end{cases}$$

rotation matrix → quaternion (Shepperd: divide by the largest component; $w \ge 0$)
$$\begin{cases} s = 2\sqrt{1 + \operatorname{tr} R},\quad q = \Big(\dfrac{s}{4}, \dfrac{R_{32} - R_{23}}{s}, \dfrac{R_{13} - R_{31}}{s}, \dfrac{R_{21} - R_{12}}{s}\Big), & \operatorname{tr} R \ge \max_i R_{ii} \\ i = \arg\max_i R_{ii},\quad s = 2\sqrt{1 + R_{ii} - R_{jj} - R_{kk}},\quad q_i = \dfrac{s}{4},\quad w = \dfrac{R_{kj} - R_{jk}}{s},\quad q_j = \dfrac{R_{ji} + R_{ij}}{s},\quad q_k = \dfrac{R_{ki} + R_{ik}}{s}, & \text{otherwise} \end{cases}$$
$$q \leftarrow \operatorname{sgn}(w)\, q / \|q\|$$

Euler angles (Z-Y-X; logs and plots only)
$$\phi = \operatorname{atan2}(R_{32}, R_{33}),\quad \theta = \arcsin(-R_{31}),\quad \psi = \operatorname{atan2}(R_{21}, R_{11})$$

heading (horizontal projection of body x)
$$\psi(R) = \operatorname{atan2}(R_{21}, R_{11}),\qquad \|(R_{11}, R_{21})\| > 0$$

wrap to $[-\pi, \pi)$
$$\operatorname{wrap}(\alpha) = \big((\alpha + \pi) \bmod 2\pi\big) - \pi$$



① Ninth-Degree Time Law $S(u)$

`trajectories/timing.py`

construction: S' has fourfold zeros at both ends
$$S'(u) = C\,u^4(1-u)^4,\qquad \int_0^1 S'\,du = C\,\frac{4!\,4!}{9!} = 1 \Rightarrow C = 630$$

ten endpoint conditions
$$S(0) = 0,\quad S(1) = 1,\qquad S^{(k)}(0) = S^{(k)}(1) = 0,\quad k = 1, \dots, 4$$

integrating gives the unique polynomial of degree ≤ 9
$$S(u) = 126u^5 - 420u^6 + 540u^7 - 315u^8 + 70u^9$$

uniqueness
$$S_1 - S_2 \text{ has fivefold zeros at } 0, 1 \Rightarrow u^5(u-1)^5 \mid S_1 - S_2,\ \deg \le 9 \Rightarrow S_1 \equiv S_2$$

normalized time
$$u = \frac{t - t_s}{T},\qquad \sigma = \begin{cases} 0, & u \le 0 \\ S(u), & 0 < u < 1 \\ 1, & u \ge 1 \end{cases}$$

symmetric evaluation (no cancellation; Horner)
$$S(u) = 1 - S(1-u),\qquad \varsigma = \min(u, 1-u)$$
$$S = \varsigma^5\big(126 + \varsigma(-420 + \varsigma(540 + \varsigma(-315 + 70\varsigma)))\big)$$

derivatives (factored, exactly zero at the ends)
$$S' = 630\,u^4(1-u)^4$$
$$S'' = 2520\,u^3(1-u)^3(1-2u)$$
$$S''' = 2520\,u^2(1-u)^2(3 - 14u + 14u^2)$$
$$S'''' = 15120\,u(1-u)(1-2u)(1 - 7u + 7u^2)$$

integral (used by cruise-law ramps (2))
$$I(u) = \int_0^u S = 21u^6 - 60u^7 + \tfrac{135}{2}u^8 - 35u^9 + 7u^{10}$$
$$I(1) = \tfrac12,\qquad I(u) = u - \tfrac12 + I(1-u)$$

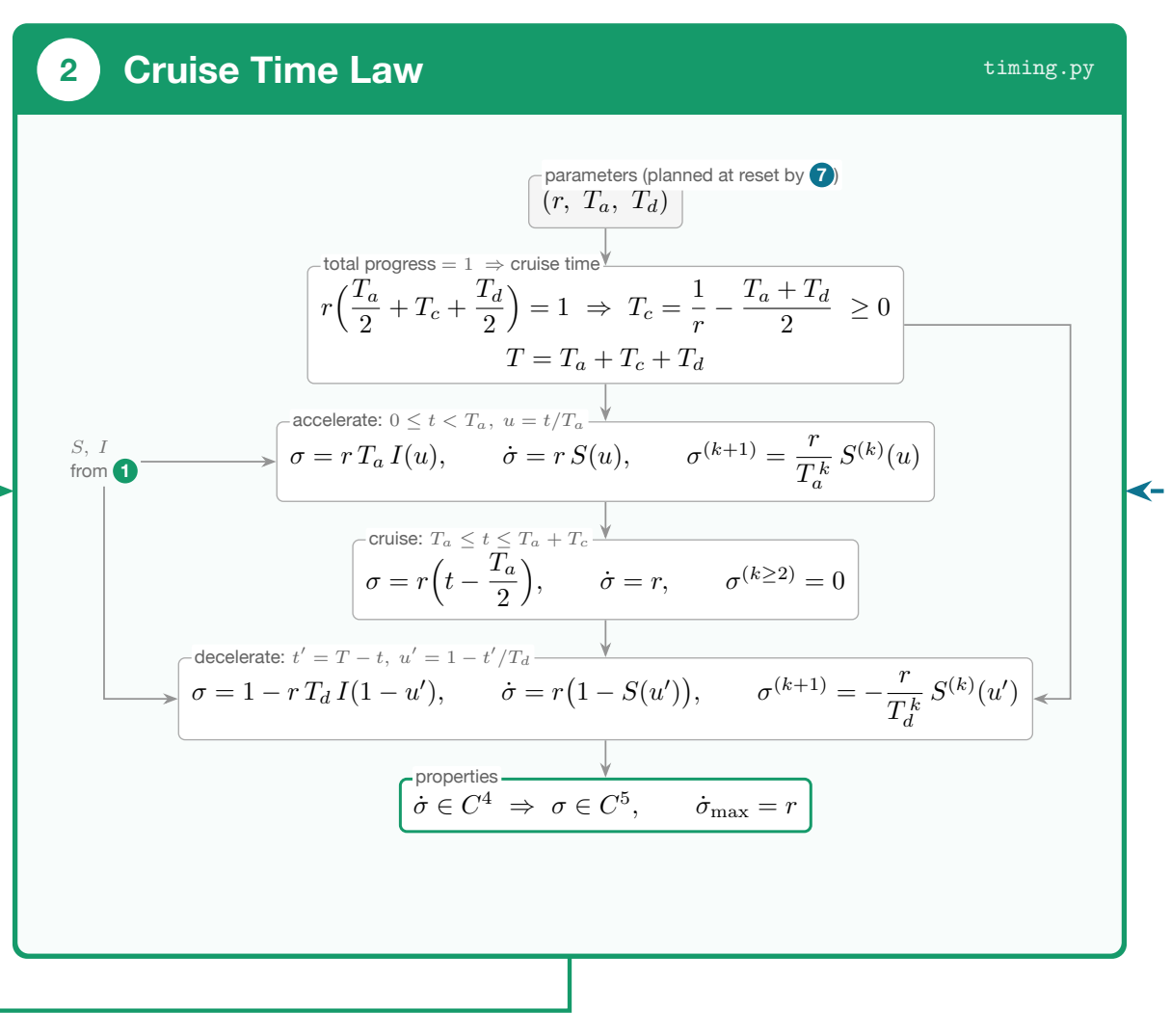
Hermite law in physical time: one factor $1/T$ per order
$$\sigma^{(k)}(t) = \frac{1}{T^k} S^{(k)}(u),\qquad k = 0, 1, 2, 3, 4$$

peak rate (midpoint)
$$\dot\sigma_{\max} = \frac{S'(\frac12)}{T} = \frac{315}{128\,T},\qquad \ddot\sigma\big(\tfrac{T}{2}\big) = 0$$

$S,\ I$

$\sigma^{(k)}$ (Hermite law)

$\sigma^{(k)}$ (cruise law)



② Cruise Time Law

`timing.py`

parameters (planned at reset by (7))
$$(r,\ T_a,\ T_d)$$

total progress = 1 ⇒ cruise time
$$r\Big(\frac{T_a}{2} + T_c + \frac{T_d}{2}\Big) = 1 \Rightarrow T_c = \frac{1}{r} - \frac{T_a + T_d}{2} \ge 0$$
$$T = T_a + T_c + T_d$$

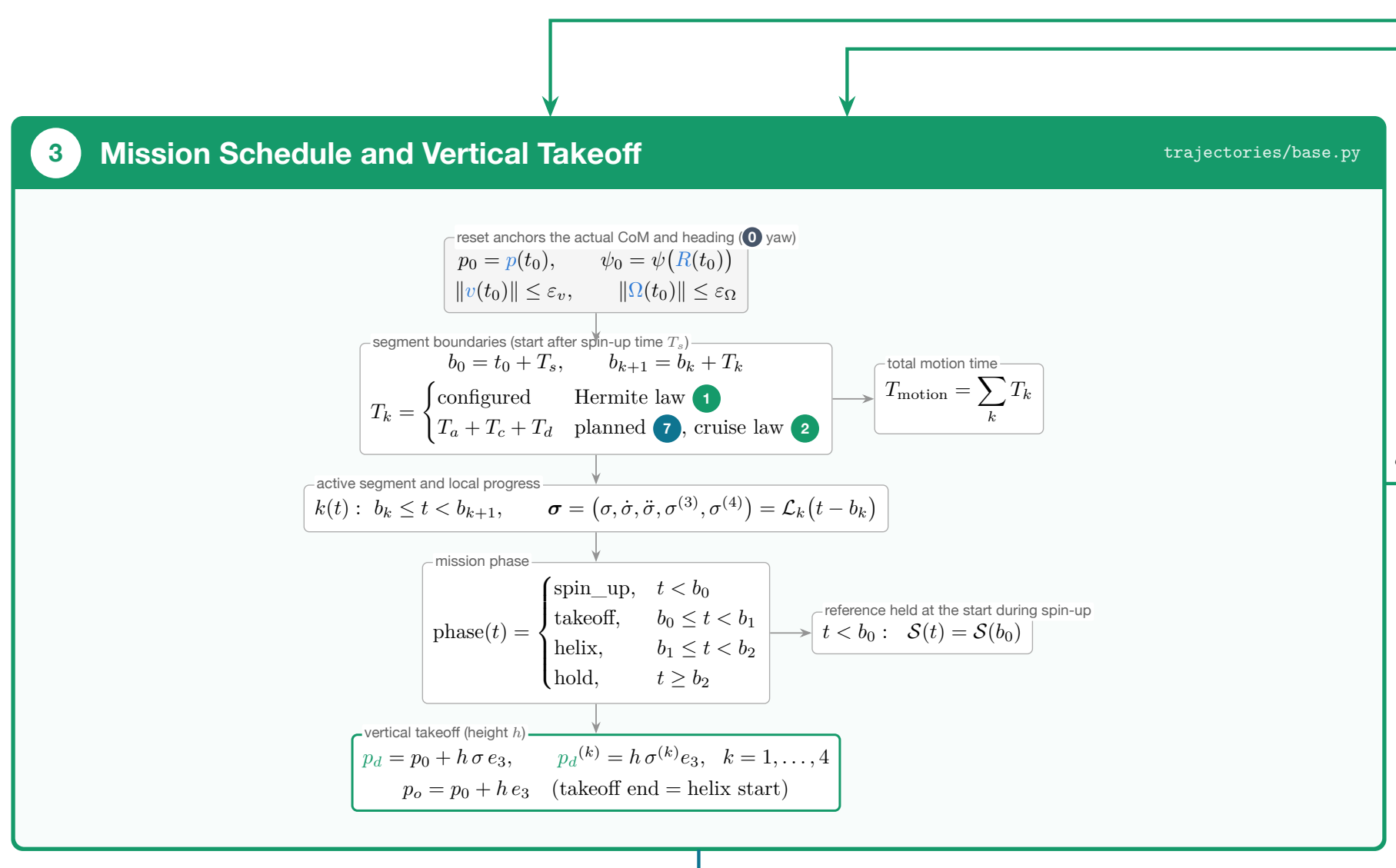
$S,\ I$ from (1)

accelerate: $0 \le t < T_a,\ u = t/T_a$
$$\sigma = r\,T_a\,I(u),\qquad \dot\sigma = r\,S(u),\qquad \sigma^{(k+1)} = \frac{r}{T_a^k} S^{(k)}(u)$$

cruise: $T_a \le t \le T_a + T_c$
$$\sigma = r\Big(t - \frac{T_a}{2}\Big),\qquad \dot\sigma = r,\qquad \sigma^{(k \ge 2)} = 0$$

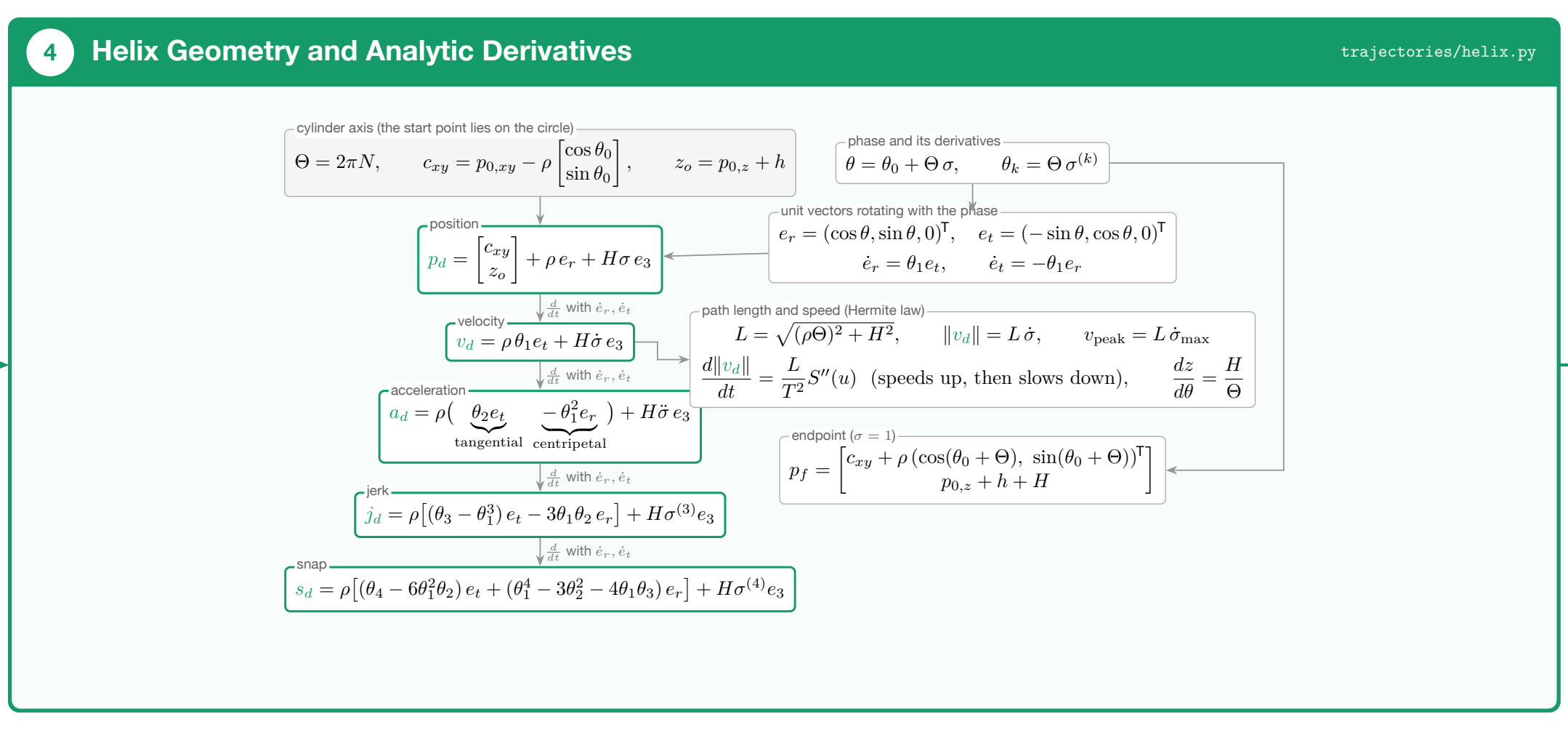
decelerate: $t' = T - t,\ u' = 1 - t'/T_d$
$$\sigma = 1 - r\,T_d\,I(1 - u'),\qquad \dot\sigma = r\big(1 - S(u')\big),\qquad \sigma^{(k+1)} = -\frac{r}{T_d^k} S^{(k)}(u')$$

properties
$$\dot\sigma \in C^4 \Rightarrow \sigma \in C^5,\qquad \dot\sigma_{\max} = r$$



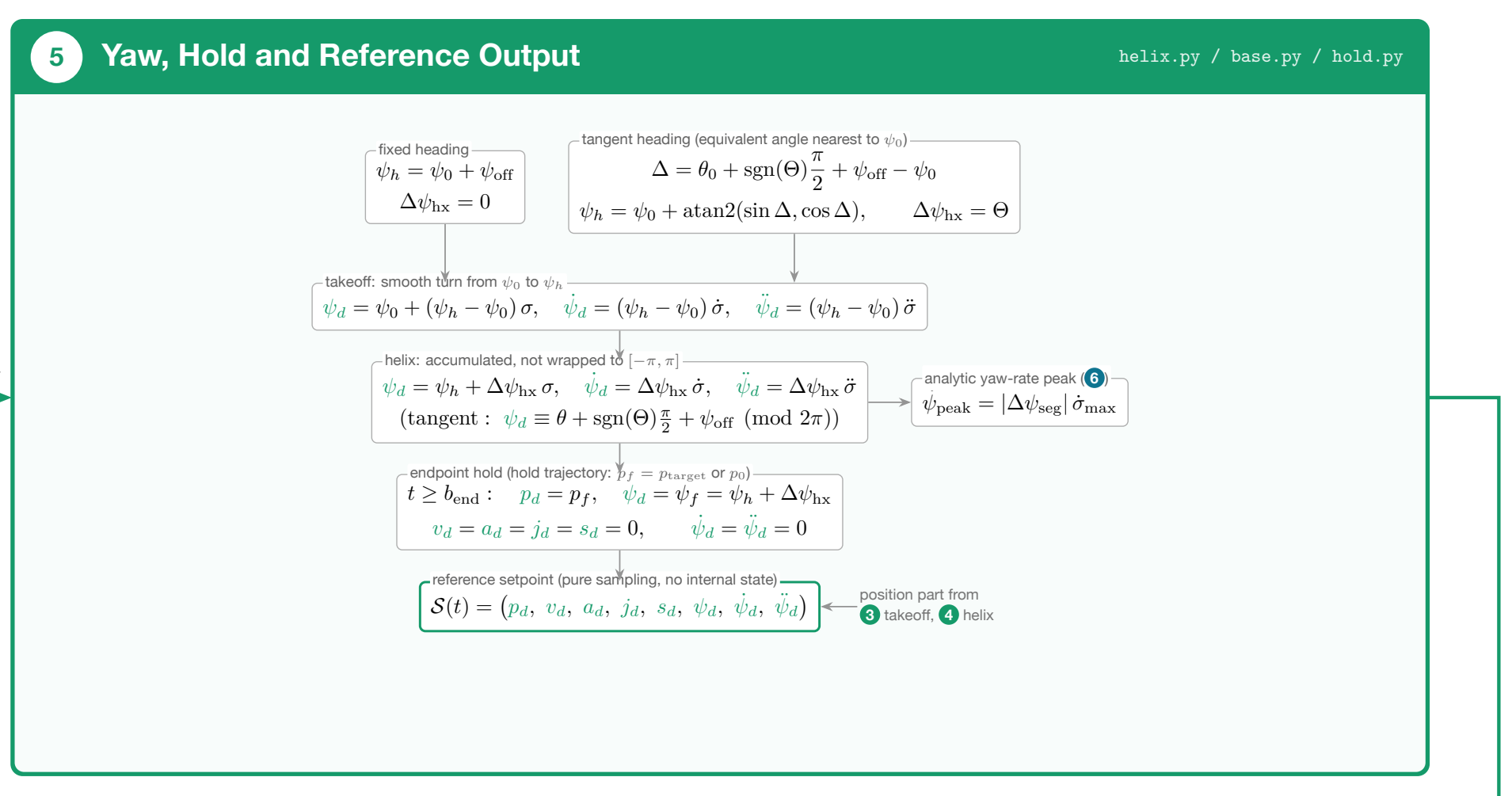
③ Mission Schedule and Vertical Takeoff

`trajectories/base.py`



④ Helix Geometry and Analytic Derivatives

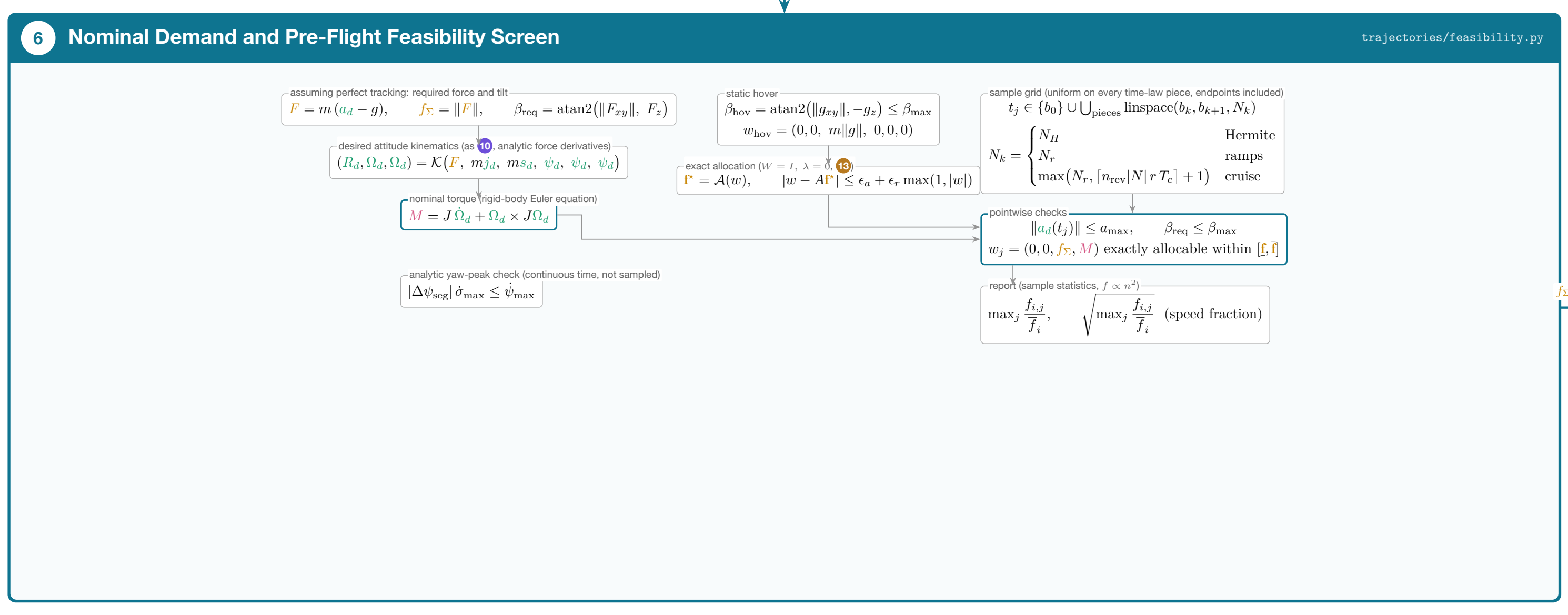
`trajectories/helix.py`



⑤ Yaw, Hold and Reference Output

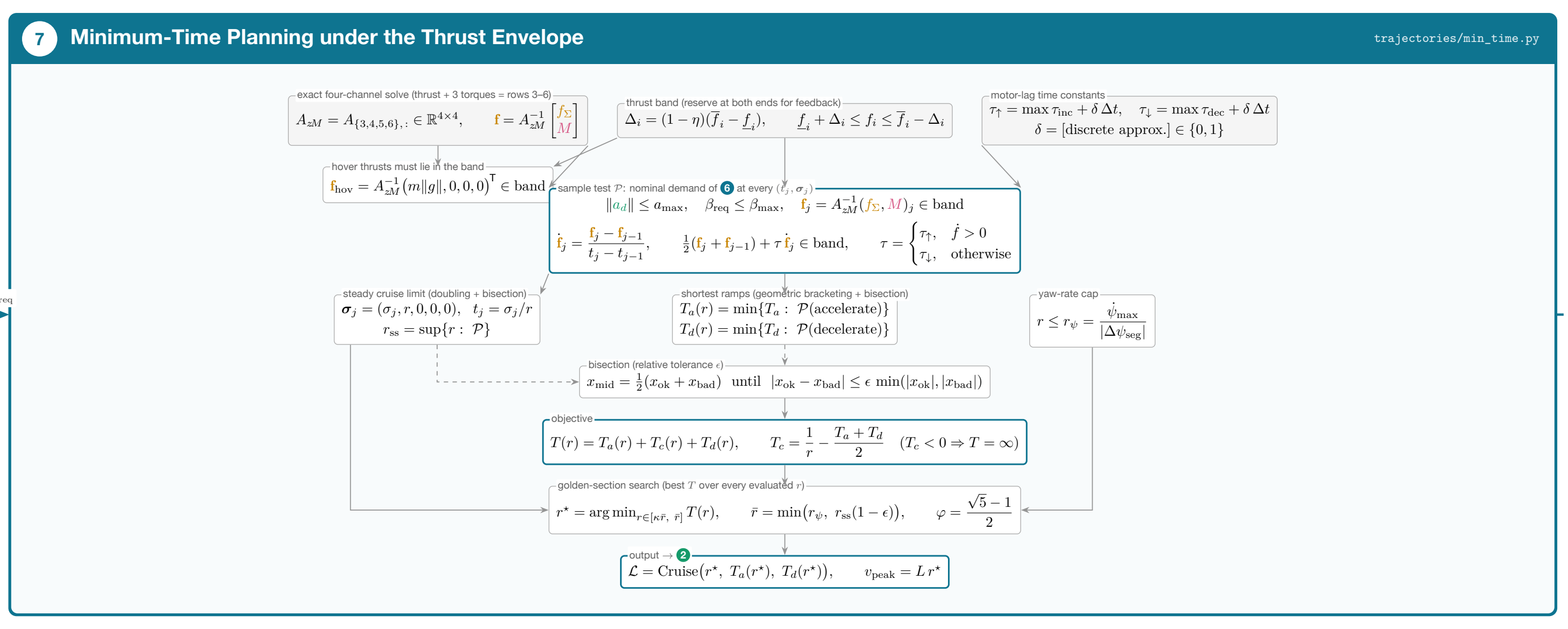
`helix.py / base.py / hold.py`

samples $\mathcal{S}(t_k)$



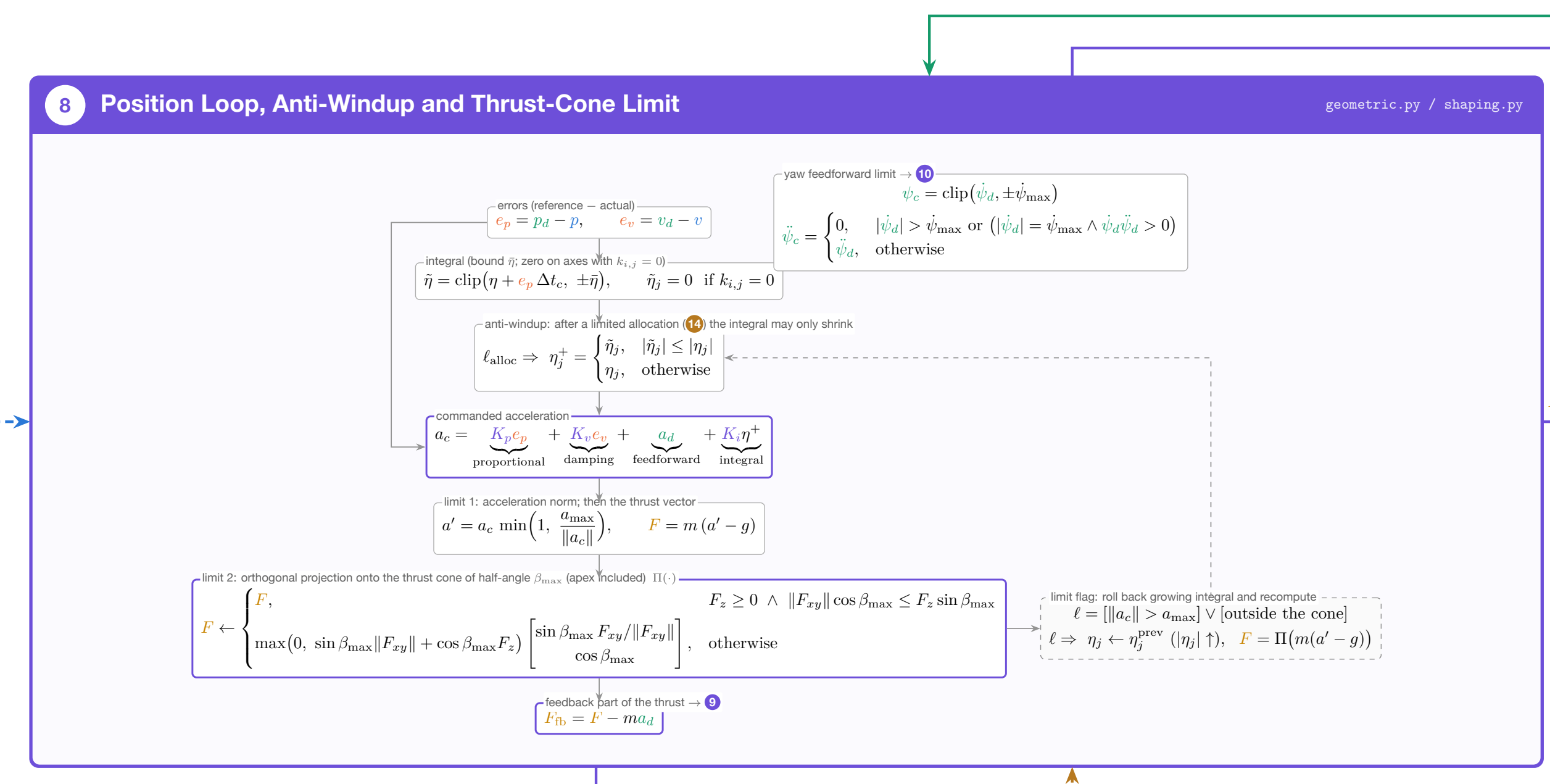
⑥ Nominal Demand and Pre-Flight Feasibility Screen

`trajectories/feasibility.py`



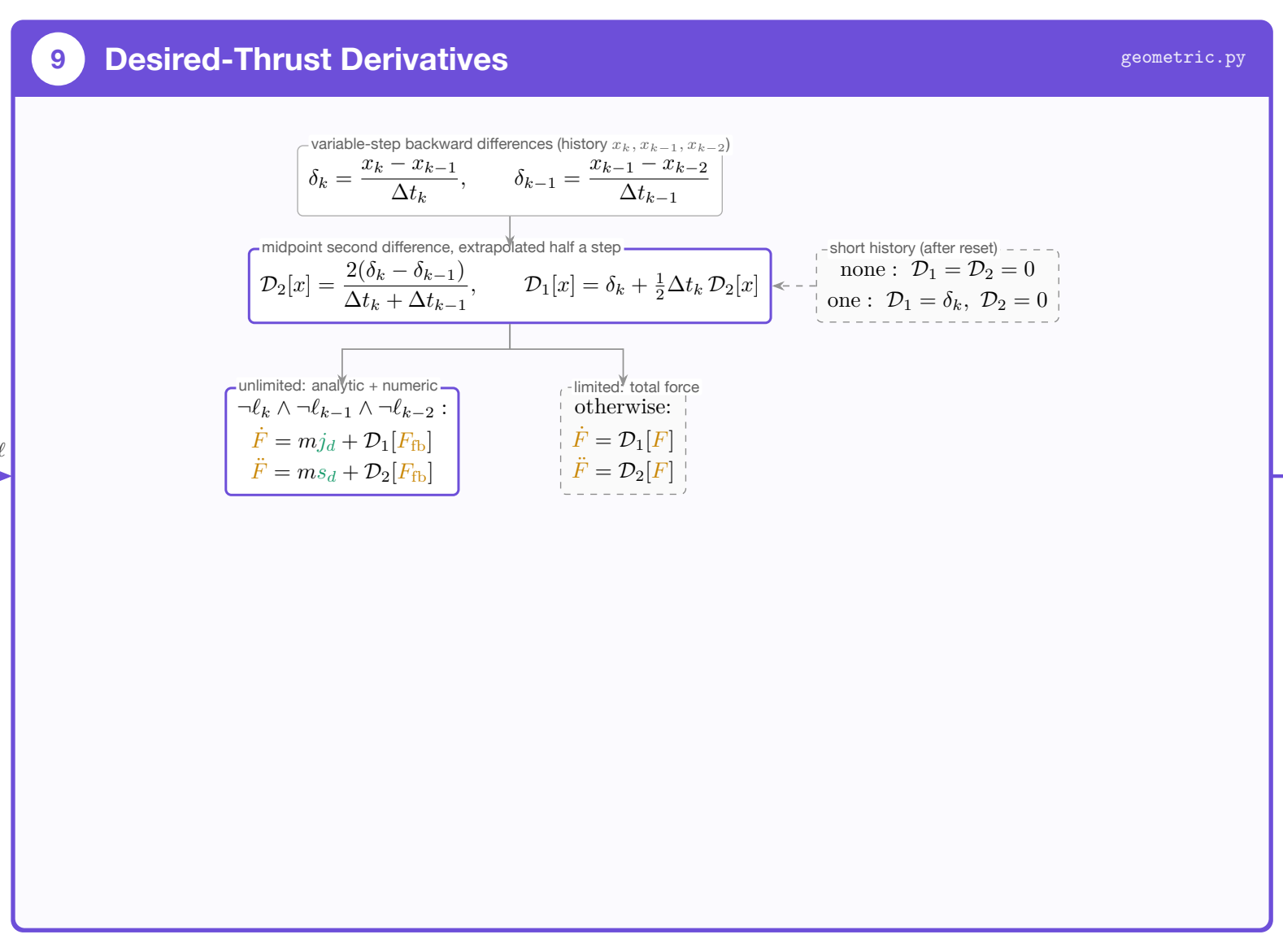
⑦ Minimum-Time Planning under the Thrust Envelope

`trajectories/min_time.py`



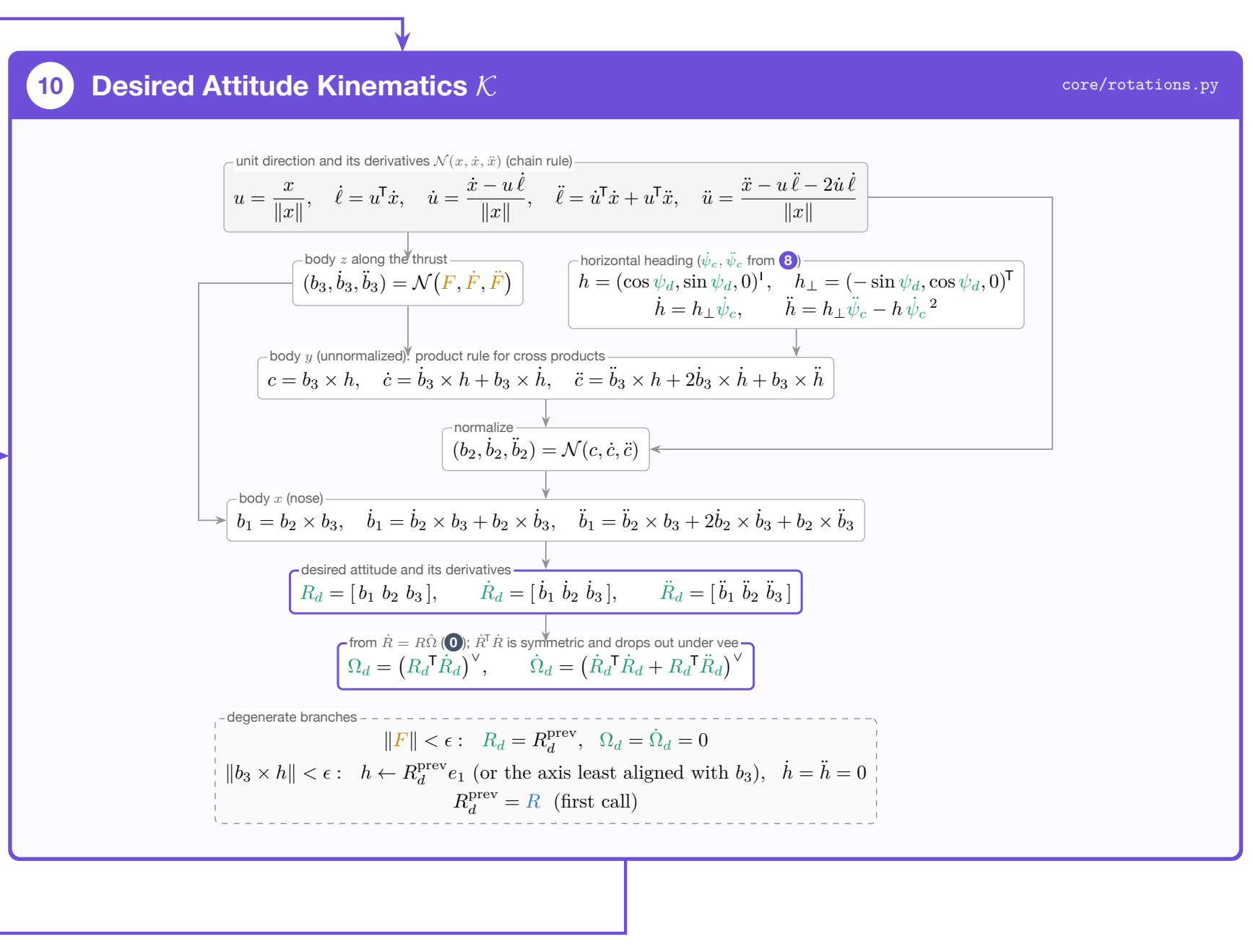
⑧ Position Loop, Anti-Windup and Thrust-Cone Limit

`geometric.py / shaping.py`



⑨ Desired-Thrust Derivatives

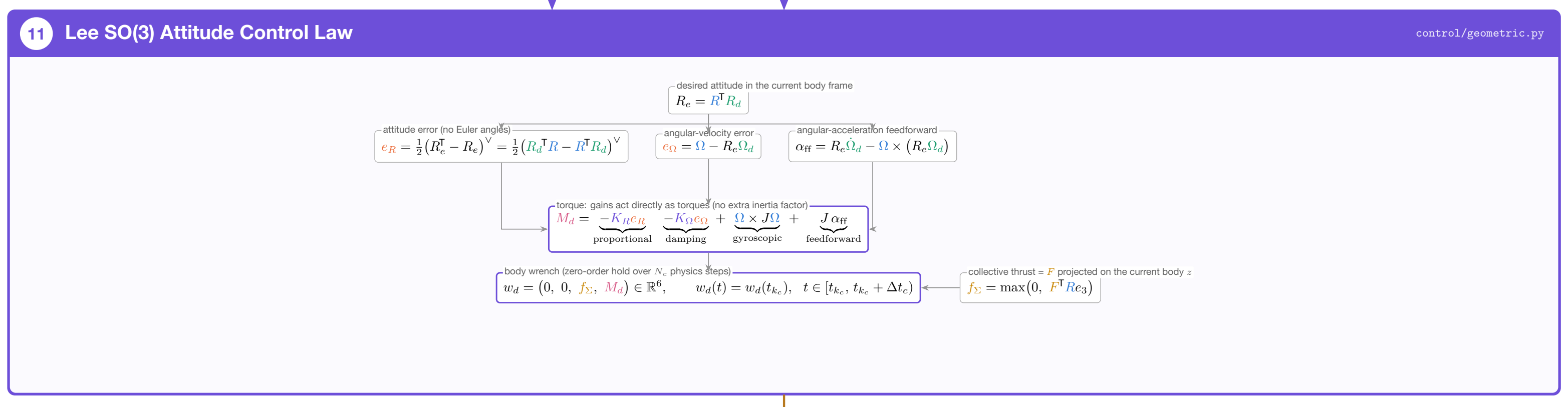
`geometric.py`



⑩ Desired Attitude Kinematics $\mathcal{K}$

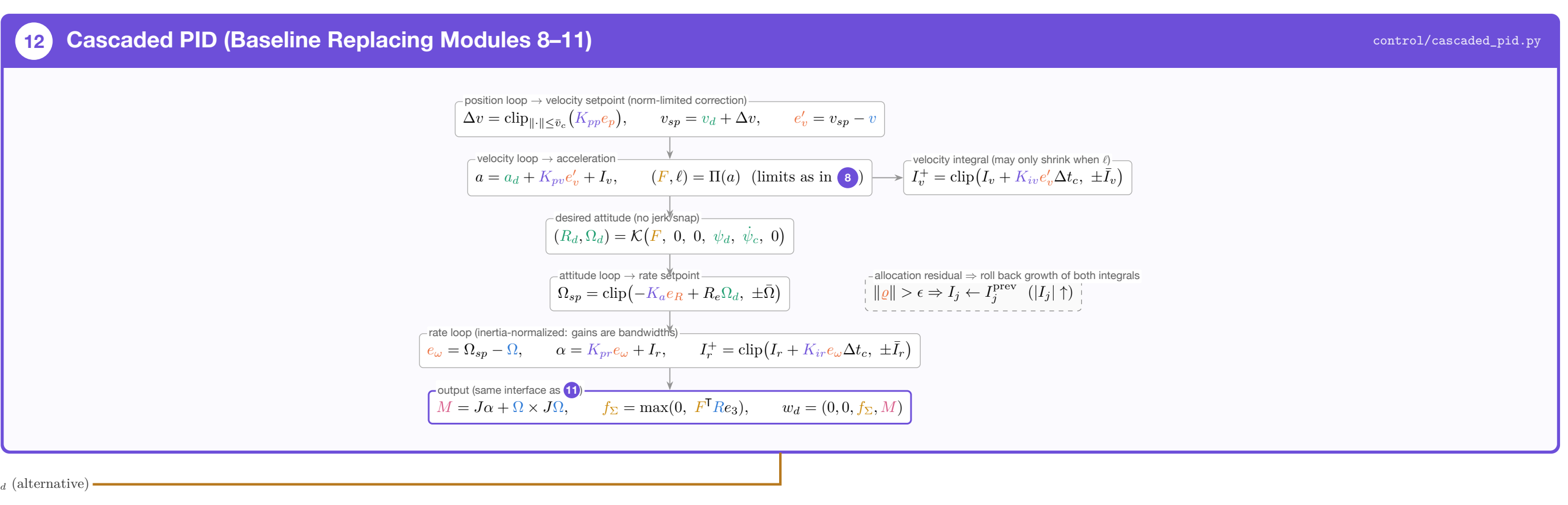
`core/rotations.py`

ℓ_{alloc} (anti-windup, next control step)



⑪ Lee SO(3) Attitude Control Law

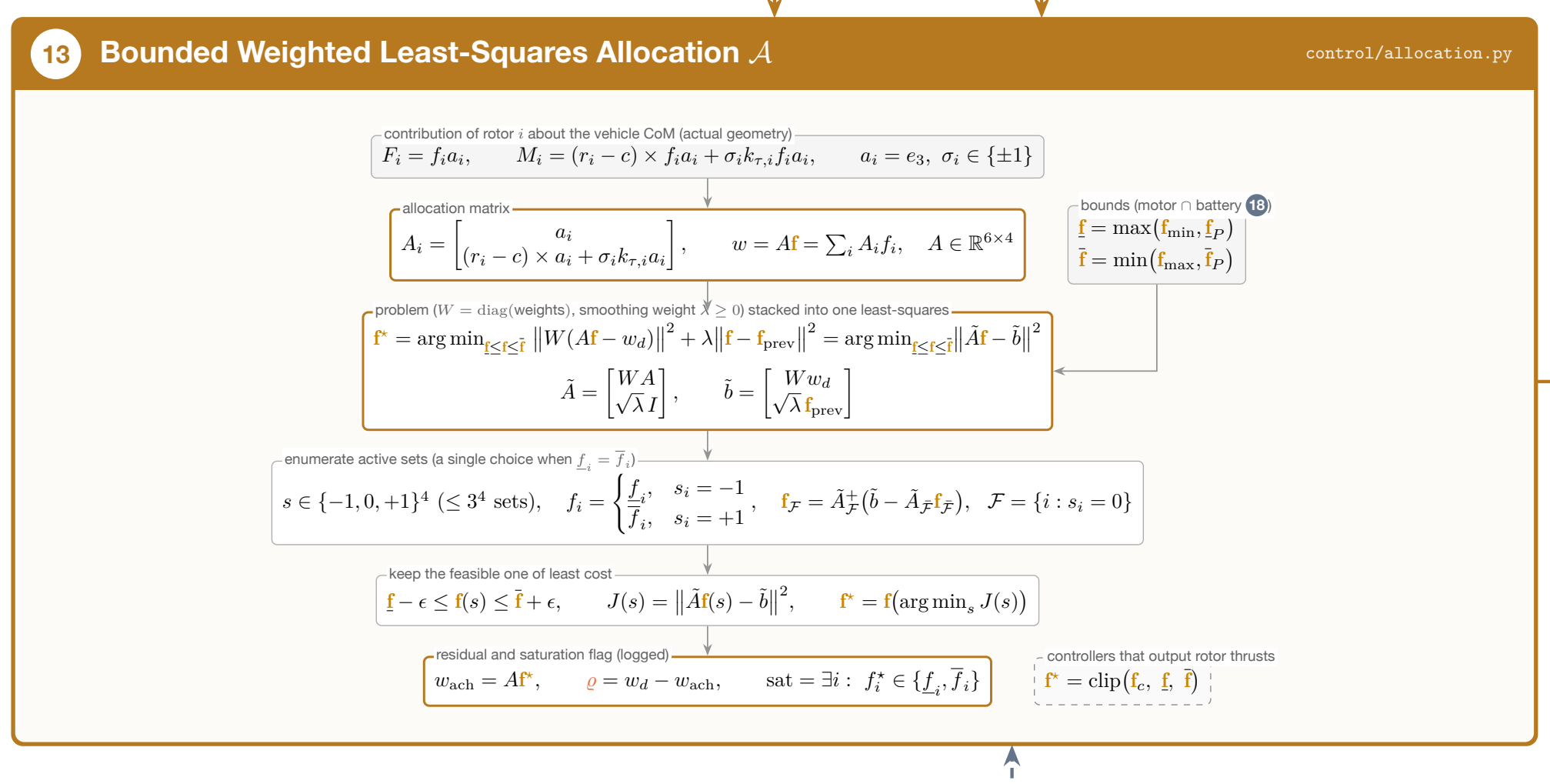
`control/geometric.py`



⑫ Cascaded PID (Baseline Replacing Modules 8–11)

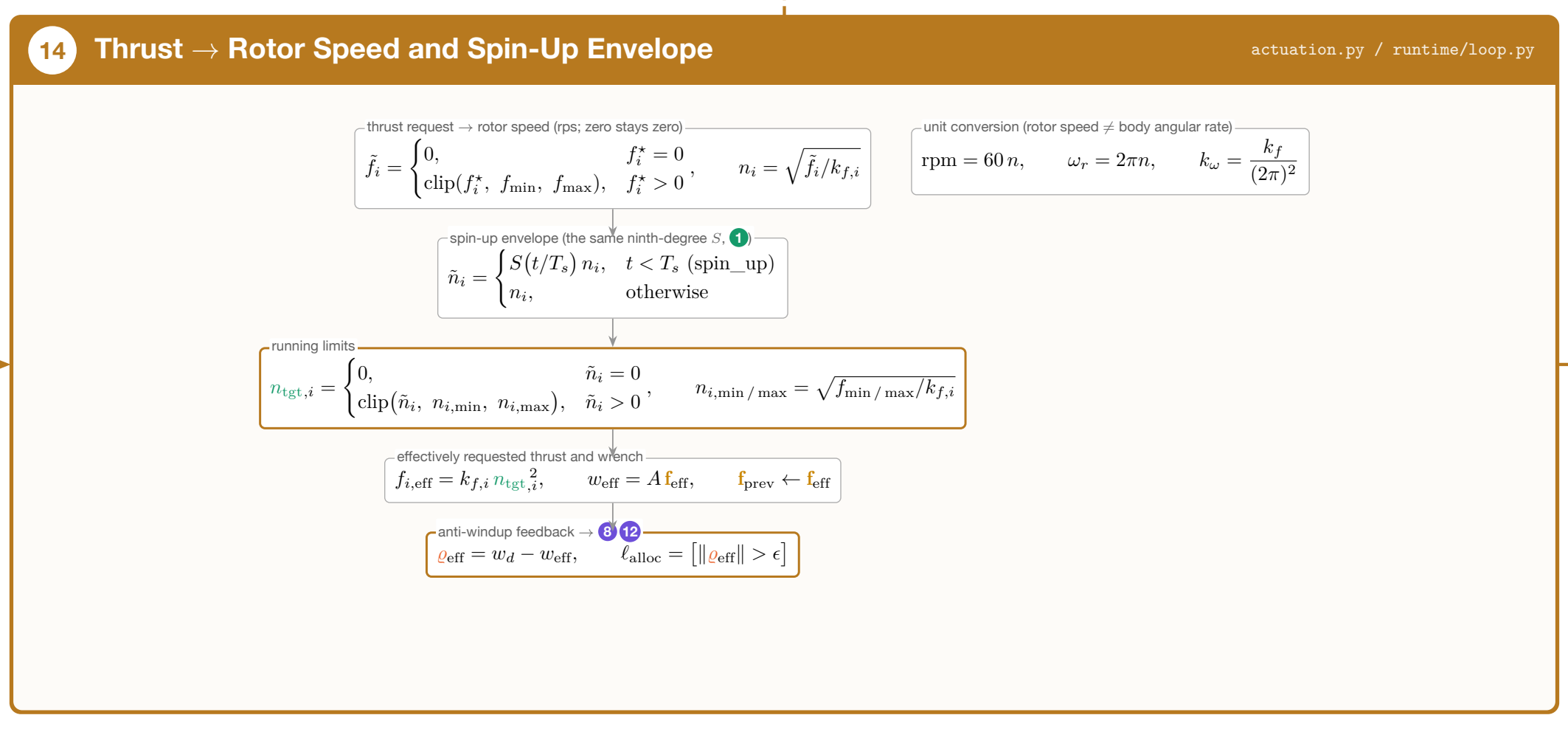
`control/cascaded_pid.py`

w_d (alternative)



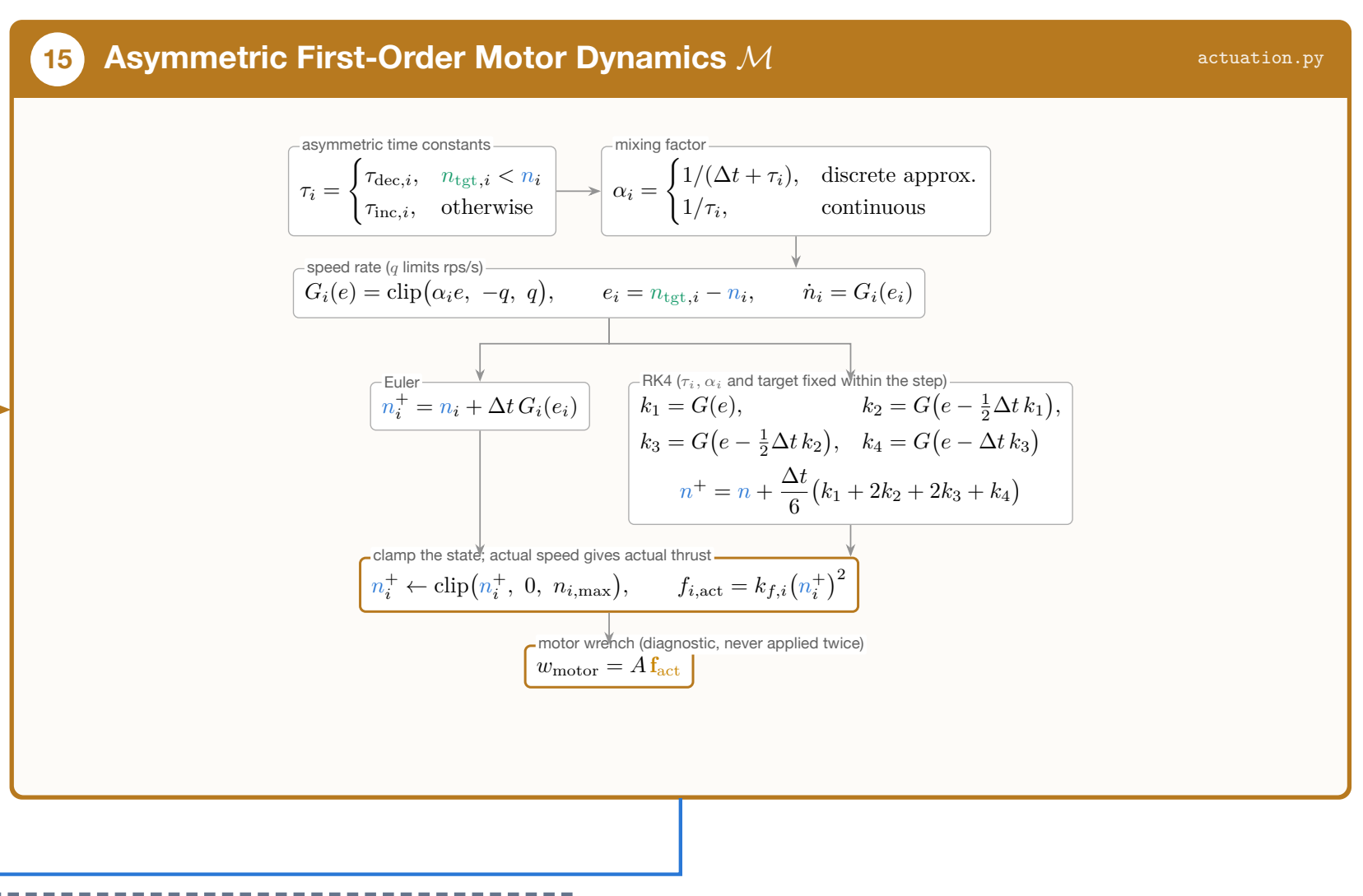
⑬ Bounded Weighted Least-Squares Allocation $\mathcal{A}$

`control/allocation.py`



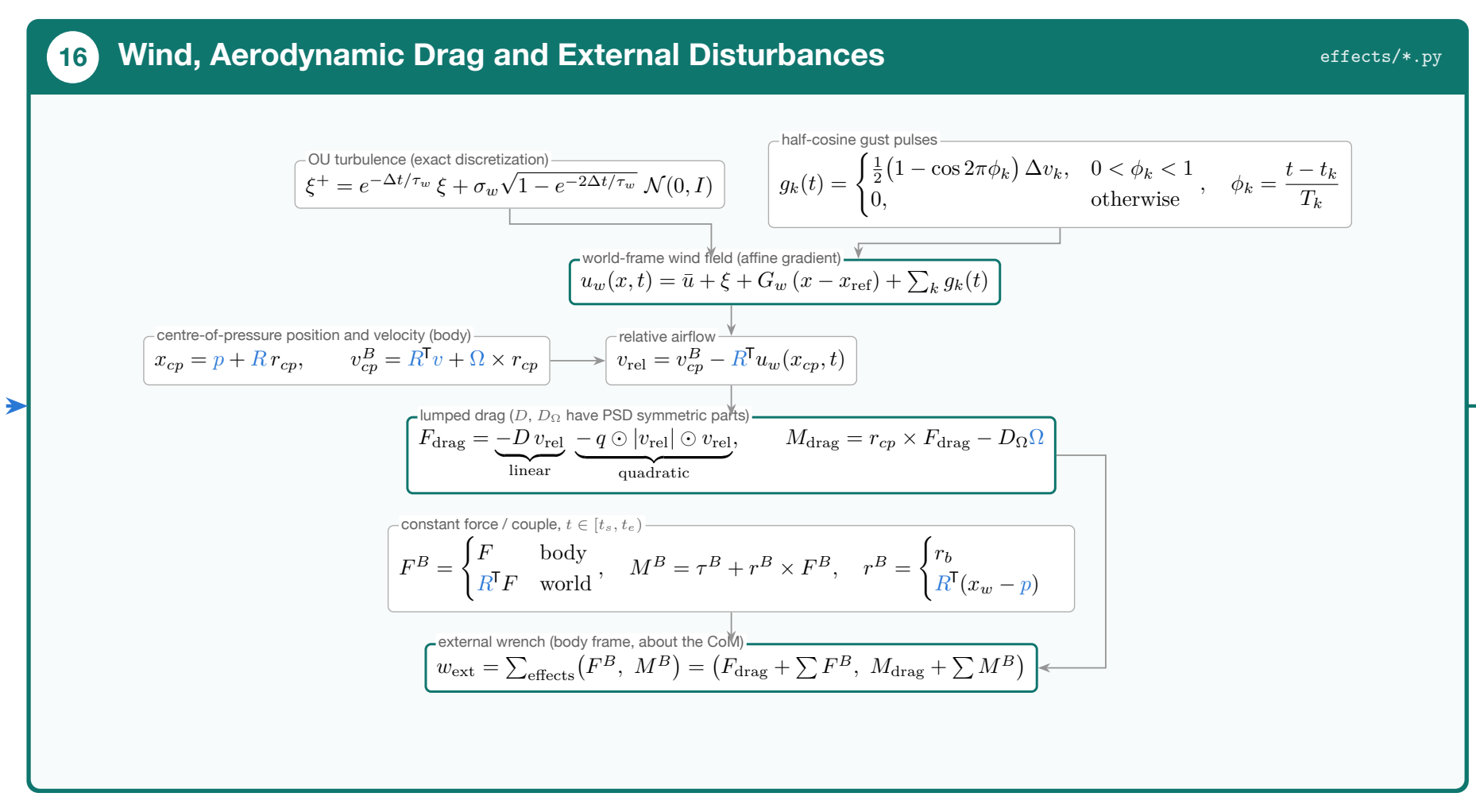
⑭ Thrust → Rotor Speed and Spin-Up Envelope

`actuation.py / runtime/loop.py`



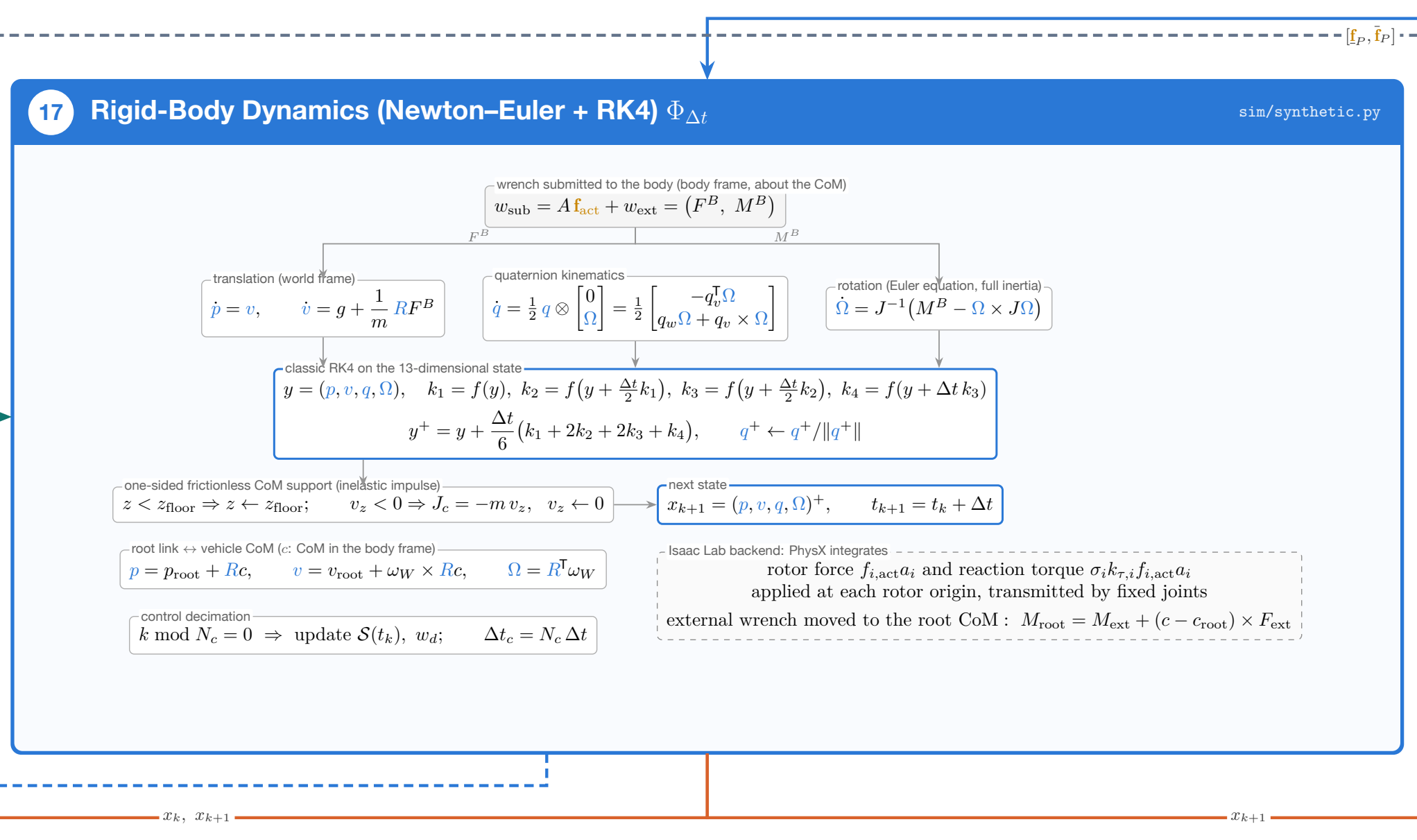
⑮ Asymmetric First-Order Motor Dynamics $\mathcal{M}$

`actuation.py`



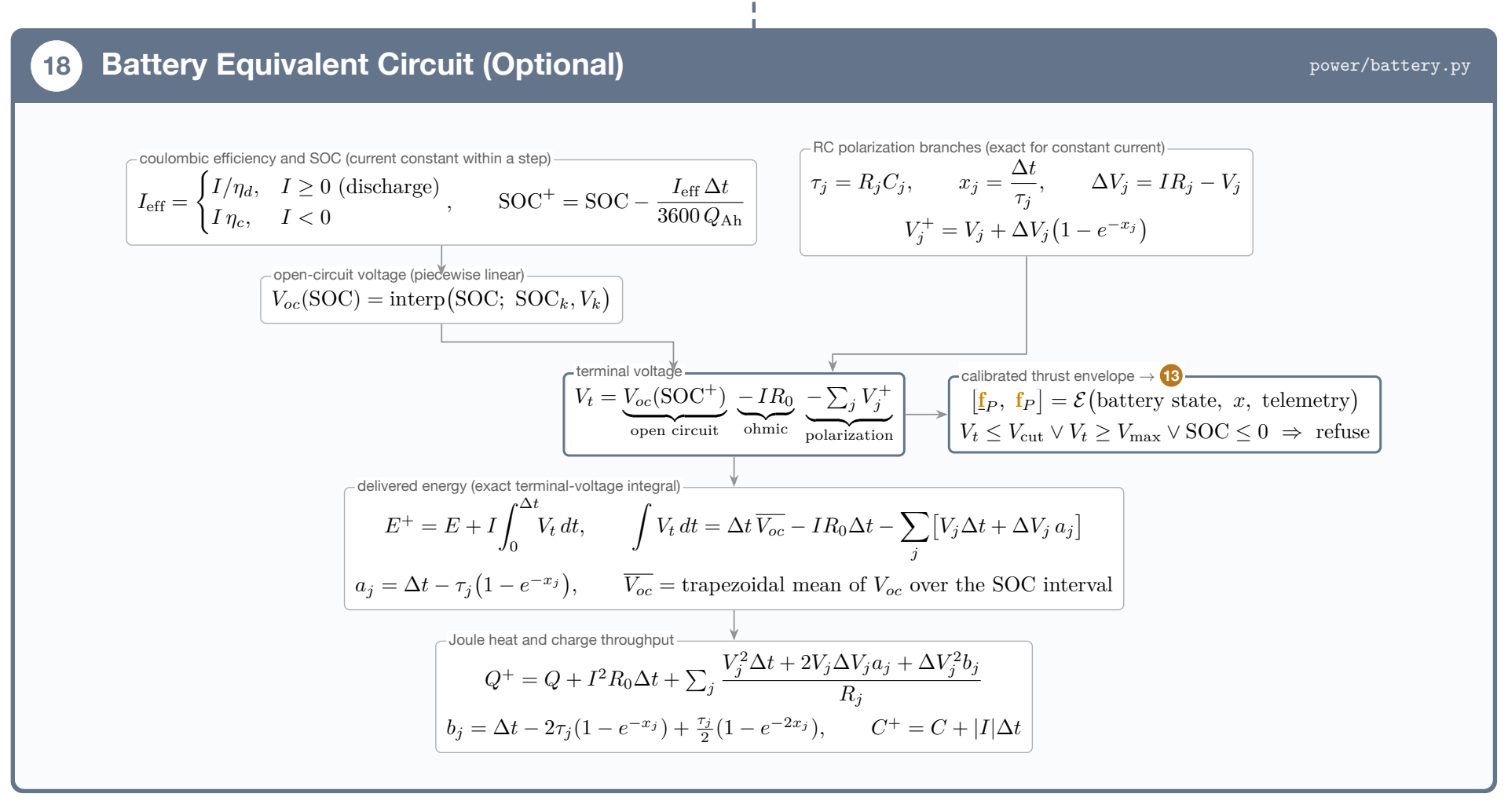
⑯ Wind, Aerodynamic Drag and External Disturbances

`effects/*.py`



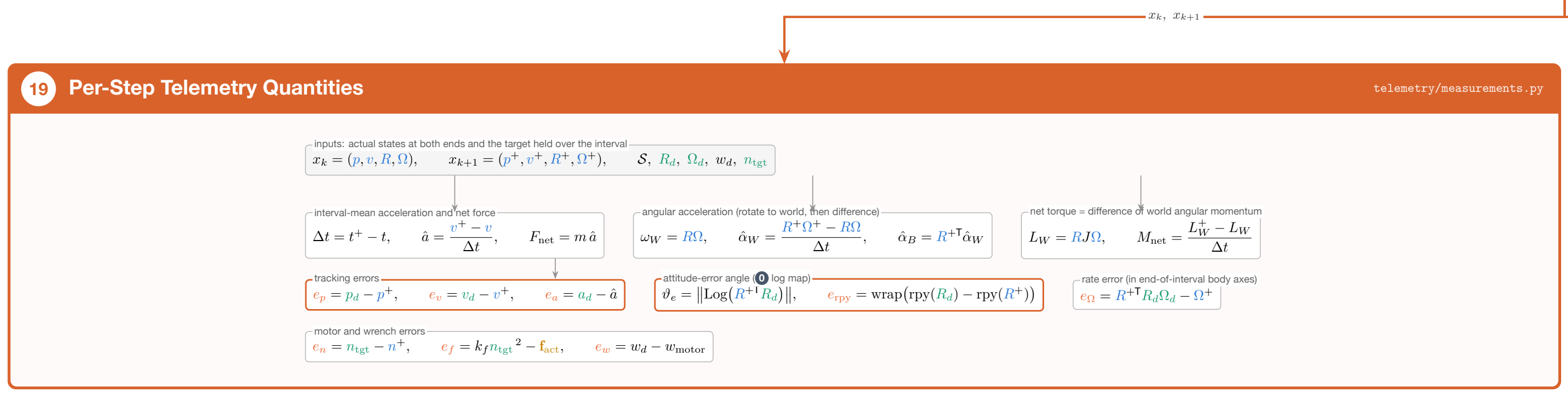
⑰ Rigid-Body Dynamics (Newton–Euler + RK4) $\Phi_{\Delta t}$

`sim/synthetic.py`



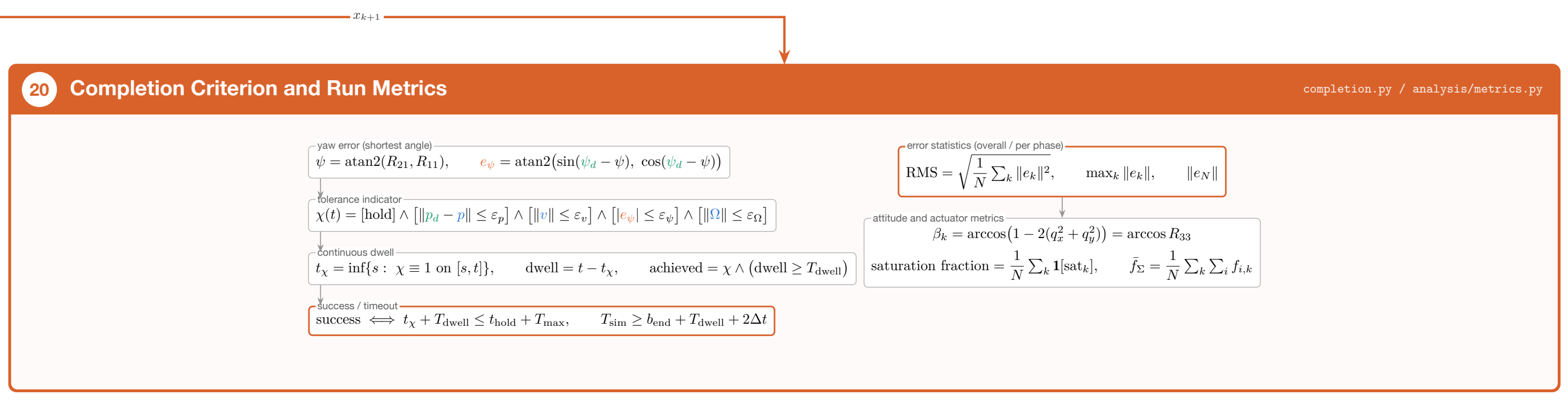
⑱ Battery Equivalent Circuit (Optional)

`power/battery.py`



⑲ Per-Step Telemetry Quantities

`telemetry/measurements.py`



⑳ Completion Criterion and Run Metrics

`completion.py / analysis/metrics.py`